

Lab 07.1: Wheel Rims

14/11/2025

Abstract

This study evaluates the fatigue life of an aluminum alloy wheel rim for an Audi A1 (2024). Using a finite element model to simulate a dynamic cornering fatigue test, the study analyzed the impact of rotating bending moments on the component.

Key findings include:

- **Stress Analysis:** the Sines equivalent stress criterion was used to combine alternate stress component
- **Peak Stress:** a maximum stress of 64.139 MPa was identified near the hub connection
- **Fatigue Life:** all calculated stresses remained below the material endurance limit
- **Safety Outcome:** the wheel rim is designed for infinite life, meeting long-term safety standards for the specified conditions

Contents

1. Introduction.....	3
1.1 Description of the problem	3
1.2 Data of selected vehicle	4
2. Computation of the Sines equivalent stress	6
3. Computation of the fatigue life on the wheel	8
4. Conclusion.....	9

1. Introduction

The goal of this laboratory is to compute the fatigue life of an aluminum alloy wheel subject to a cornering fatigue test.

1.1 Description of the problem

During the design phase, a wheel undergoes many different tests to make sure that some safety and performance standards are met. In this case, safety tests are taken into consideration.

Two types of safety tests can be identified:

- Fatigue tests:
 - Types of fatigue tests:
 - Lateral fatigue tests
 - Radial fatigue tests
 - Combined/biaxial fatigue tests
- Impact tests:
 - Types of impact tests:
 - Lateral impact tests
 - Radial impact tests

Among all the tests described previously, the one that is going to be discussed and analyzed in this laboratory is the **dynamic cornering (lateral) fatigue test**.

To perform this assessment in a laboratory, a wheel is mounted on a machine that applies a rotating bending moment to the coupling surface of the wheel and the hub (Figure 1):

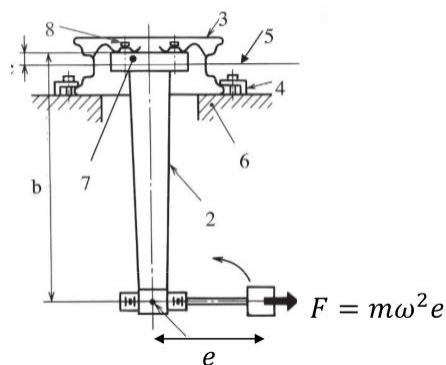


Figure 1

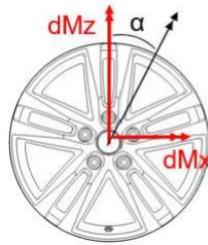
The applied bending moment is

$$M = (\mu R + d) * F_v * S$$

- $M \rightarrow$ bending moment [Nm]
- $\mu \rightarrow$ friction coefficient between tire and road (0.7 is used)
- $R \rightarrow$ tire radius [m]
- $d \rightarrow$ inset/outset of the wheel [m]
- $F_v \rightarrow$ maximum vertical static load [N]
- $S \rightarrow$ accelerated test factor (2 is used to increase the amplitude of the load and reduce the number of cycles that need to be applied)

The main concern with real testing in laboratory is related to the fact that it is a time consuming and expensive procedure; for this reason, digital twins/models are used. In this case, a finite element model has been used.

The bending moment is computed as the superposition of two orthogonal unitary bending moments $dM_z, dM_x \rightarrow$ the two stress states are combined and scaled to simulate the stress state due to the rotating bending moment that would have been applied during a real test:



$$\sigma(\alpha) = M(\sigma_{dM_z} \cos \alpha + \sigma_{dM_x} \sin \alpha)$$

- σ is the stress state due to the rotating bending moment
- M is the bending moment that is being simulated
- σ_{dM_z} is the stress state due to the unitary bending moment about the z axis
- σ_{dM_x} is the stress state due to the unitary bending moment about the x axis
- α is the angular coordinate

The assignment provides the two stress states due to the orthogonal bending moments; their superposition has been implemented successively.

1.2 Data of selected vehicle

The vehicle chosen is the Audi A1 from 2024, its wheels characteristics are the following:

- **Wheel** : 195/55R16 91 V

- Wheel radius : 0.3105 m
- Maximum vertical static load supported by the tire: 615 kg (91 index)
- **Rim** : 6.5Jx16 ET40
 - Outset : 40e-3 m

2. Computation of the Sines equivalent stress

In order to evaluate the fatigue life of the component, the Sines equivalent stress was evaluated. This criterion is based on the evaluation of the alternate components of the stresses (both normal and tangential) in the three principal directions. These are then combined as follows

$$\sigma_s = \sqrt{\sigma_{x,a}^2 + \sigma_{y,a}^2 + \sigma_{z,a}^2 - (\sigma_{x,a}\sigma_{y,a} + \sigma_{x,a}\sigma_{z,a} + \sigma_{y,a}\sigma_{z,a}) + 3(\tau_{xy,a}^2 + \tau_{xz,a}^2 + \tau_{yz,a}^2)}$$

The Siens stress was evaluated for each point of the wheel rim and the results are reported in Figure 2.

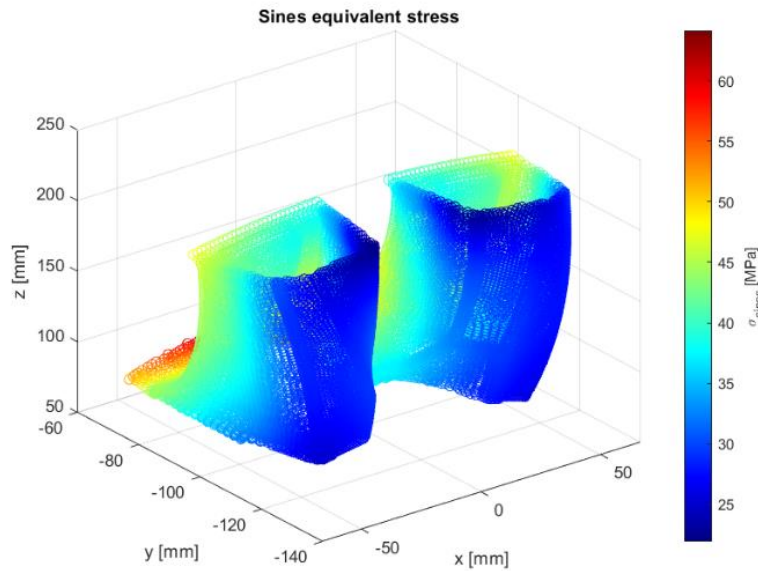


Figure 2

As far as the maximum Sines equivalent stress is concerned, it is reported in Figure 3 and it is equal to $\sigma_s = 64,139MPa$. As expected, the most stressed part of the rim is the one near the connection with the hub, since all the main loads acting on the wheel are transferred to the suspension and the body of the car there. Moreover, being the rim clamped to the hub, all the components of forces and moment cause a stress concentration there.

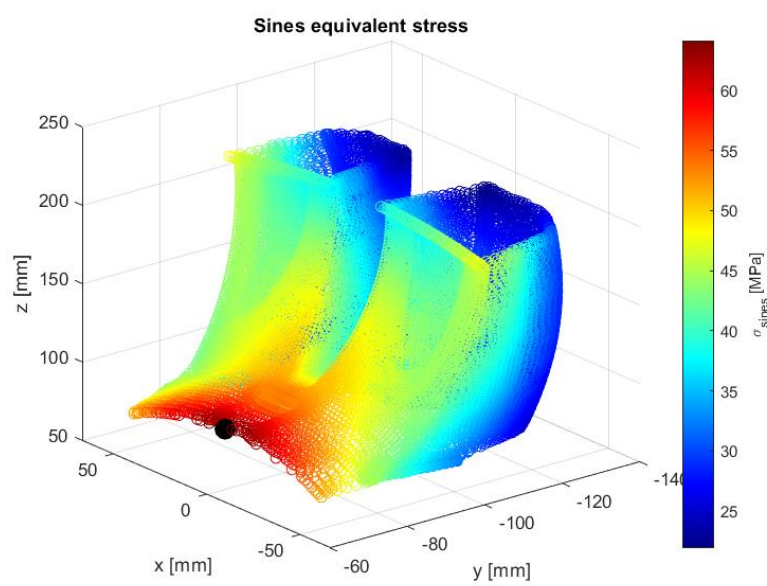


Figure 3

3. Computation of the fatigue life on the wheel

The Wohler's curve is expressed by a straight line in the $\log(\sigma) - \log(N)$ plane:

$$\log(\sigma_{lim}) = \log(\sigma'_f) + b * \log(2N_f)$$

To estimate the number of cycle N_f , by logarithmic properties we obtain:

$$N_f = \frac{1}{2} * \sqrt[b]{\left(\frac{\sigma_{lim}}{\sigma'_f}\right)}$$

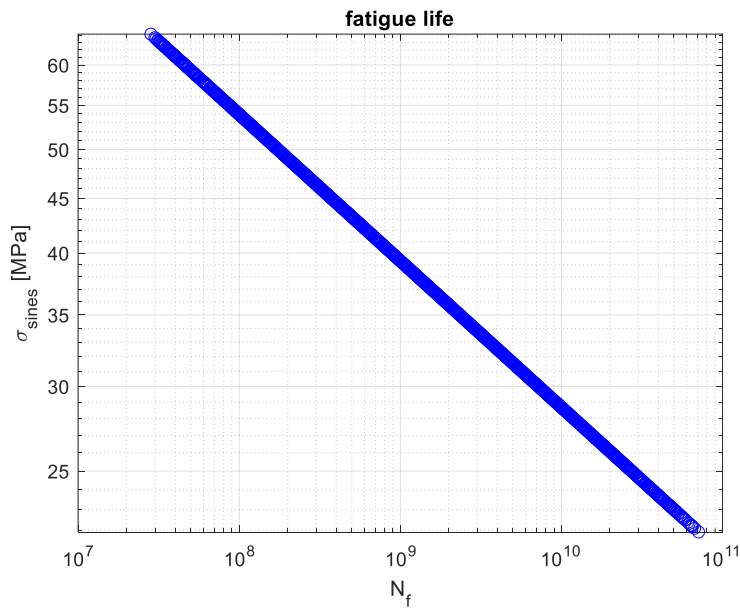


Figure 4

As illustrated in Figure 4, all applied load cases result in stress levels below the endurance limit (10^6). Consequently, the component is designed for infinite life under these conditions, ensuring its long-term safety.

4. Conclusion

This study evaluated the fatigue life of an aluminum alloy wheel rim for a 2024 Audi A1 by simulating a dynamic cornering fatigue test. With a finite element model, the analysis effectively bypassed the high costs and time requirements of physical laboratory testing while maintaining high accuracy.

The investigation yielded the following results:

- **Stress Analysis:** the Sines equivalent stress criterion was successfully used to combine alternate stress components across the rim
- **Critical Points:** a maximum stress of 64.139 MPa was recorded near the hub connection. This concentration is expected as it is the primary point where loads are transferred to the vehicle's body and suspension
- **Safety and Longevity:** all identified stress values remained below the material endurance limit

Consequently, the wheel rim is categorized as having infinite fatigue life under the specified conditions. This confirms that the design meets the rigorous safety and performance standards required for long-term operational safety.